

List of Sums

Time limit: 1.00 s

Memory limit: 512 MB

List A consists of n positive integers, and list B contains the sum of each element pair of list A .
For example, if $A = [1, 2, 3]$, then $B = [3, 4, 5]$, and if $A = [1, 3, 3, 3]$, then $B = [4, 4, 4, 6, 6, 6]$.
Given list B , your task is to reconstruct list A .

Input

The first input line has an integer n : the size of list A .

The next line has $\frac{n(n-1)}{2}$ integers: the contents of list B .

You can assume that there is a list A that corresponds to the input, and each value in A is between $1 \dots k$.

Output

Print n integers: the contents of list A .

You can print the values in any order. If there are more than one solution, you can print any of them.

Constraints

- $3 \leq n \leq 100$
- $1 \leq k \leq 10^9$

Example

Input	Output
4 4 4 4 6 6 6	1 3 3 3

Explanation: In this case list A can be either $[1, 3, 3, 3]$ or $[2, 2, 2, 4]$ and both solutions are accepted.